

MELEK BOUABID

Computer Engineering Student — Data Science & Artificial Intelligence

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Profile

5th-year Computer Engineering student at Tek-Up, specialized in Data Science & AI. Skilled across the full ML/AI lifecycle (data preparation, feature engineering, XGBoost/Random Forest, MLflow, FastAPI deployment). Hands-on experience in NLP and RAG (LangChain, vector databases). Solid fullstack foundation (React, Django, FastAPI, Docker).

Education

Computer Engineering Degree — Data Science & AI

2024 - 2027

Tek-Up, Tunisia — 5th year (in progress)

Technical Skills

- **Languages:** Python, SQL, JavaScript, Java, C, C++, C#, HTML, CSS
- **Web/Backend:** Flask, FastAPI, Django, Spring Boot, .NET, Node.js, Flutter
- **ML/DL:** Scikit-learn, XGBoost, Random Forest, TensorFlow, PyTorch, Keras, Pandas, NumPy
- **Generative AI / RAG:** LangChain, vector databases, semantic search, prompt engineering
- **Frontend:** React, TypeScript, Tailwind CSS
- **Data & Tools:** PySpark, MySQL, PostgreSQL, MongoDB, Neo4j, Git, Docker, MLflow, Power BI

Professional Experience

Data Science / AI Intern — STIET-Philips

2026 (2 months)

- Contributed to a smart CMMS POC (Laravel 12) for tracking medical installations (MRI and CathLab rooms), structuring operational, technical, financial and documentary data across the installation lifecycle.
- Implemented a conversational assistant based on a documentary RAG architecture (document indexing and chunking, relevant passage retrieval, answer generation via the Gemini API) for administrators and managers.
- Designed decision-support scoring models: equipment risk score (warranty, age, incidents, complaints, maintenance history), team performance score, and statistical prediction of installation duration from archived data.
- Took part in designing an interactive installation-readiness graph (equipment, components, safety elements, documents) with status indicators, and in setting up caching and re-indexing mechanisms.

Mobile Development Intern (Flutter) — Advanced Technologies Software

2025 (1 month)

- Built a Flutter mobile app for Ray-Ban enabling in-store barcode/QR code scanning of a product and real-time stock availability checks.
- Designed the scanning module (camera integration and barcode/QR reader) and the communication logic with the product-availability backend, and took part in testing and bug fixing before release.

Projects

CSIS — Criminal Study Intelligent System

2025

- Trained and compared supervised ML models (XGBoost, Random Forest) to detect social risk among young people, with data cleaning, feature engineering and NLP for text content analysis.
- Designed an intelligent legal assistant based on a RAG architecture (LangChain + vector database): ingestion and chunking of Tunisian law texts, semantic search and context-grounded answer generation reducing hallucination risk.
- Presented risk scores, NLP results and the RAG assistant through a Streamlit interface and a Power BI dashboard.

Riot Games Analytics Platform

2026

- Built an end-to-end ML platform on League of Legends data: classification/regression models to predict match outcomes and player rank, and an anomaly detection model to flag suspicious accounts (smurfs).
- Built the full pipeline (data collection via the Riot Games API with caching and retry/backoff, feature engineering, training with Scikit-learn), experiment tracking with MLflow, exposed via a FastAPI backend consumed by a React frontend.

The Room — Escape Game Booking Website

- Built a public online booking website (React) and an admin dashboard, with an Express + PostgreSQL backend containerized with Docker, real-time notifications (SSE) and PDF/Excel data export.

E-Bike Rental Platform

- Designed a peer-to-peer e-bike rental marketplace using Django REST Framework for the backend and React/TypeScript for the frontend, including JWT authentication and Stripe payment integration.

Generator Data Crime

- Built a React/Vite application generating synthetic criminology datasets (JSON/CSV) with realistic statistical correlations, used to test data processing pipelines and analytical dashboards.

Languages

Arabic: Native | French: Fluent | English: Fluent